Heat-Kernel Context Compression: Continuous-Time Diffusion for Token-Efficient Language-Model Inference

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Abstract

Most of what a large language model costs at inference time is tokens, and a great many of those tokens are redundant. Retrieval and long-context prompts are the worst offenders: they pack the prefill with near-duplicate text and pay full input rate for context the model never really uses. We attack this with *Heat-Kernel Context Compression* (HKCC). The idea is to place the context tokens on a similarity graph and run the graph heat equation over their embeddings. Heat flow erases high-frequency detail first, so how far a token drifts under diffusion—its *heat residual* $\rho_i(\tau) = \left\| x_i - [e^{-\tau\mathcal{L}}X]_i \right\|$—measures how much it stands apart from its neighbors. Distinctive tokens become anchors; the rest are folded into diffused, multiplicity-weighted centroids. The merge is the part that matters: deleting redundant tokens is ruinous, while re-expressing their mass costs almost nothing. A single diffusion time τ sets how hard the method compresses, and because the diffusion operator is a Markov average, every compressed token stays inside the convex hull of the originals—nothing is invented along the way.

HKCC is training-free and runs in linear time through a Chebyshev approximation that never forms the matrix exponential. We also derive a dual "budget-as-heat" rule that spreads a fixed token budget across a multi-stage pipeline until a token buys the same marginal value everywhere—the water-filling optimum—reached by purely local updates at a rate set by the pipeline's algebraic connectivity. In ten synthetic experiments HKCC holds 0.999 attention fidelity at 8% token retention, where pruning by the same score collapses to 0.34. An ablation pins the credit on the diffused merge rather than the score. The usable region is a broad plateau, not a tuned point; diffusion cost is linear where the exact operator is cubic; and the savings stack with prompt caching and batching to roughly 70× at the operating point. We finish with a protocol for a real-LLM evaluation and an honest account of what is and isn't new here.

Keywords: token compression, graph heat equation, spectral graph theory, efficient inference, retrieval-augmented generation, context engineering.

1 Introduction

If you run LLMs in production, your bill is mostly tokens. Input tokens are billed on every prefill; output tokens come out one at a time and, on the major commercial models, run something like

3–8× the input rate. In retrieval-augmented generation (RAG) and long-context systems the prefill is where the money goes—a pipeline pastes in several long documents when a single paragraph holds the answer, and pays for all of it on every call. Most of that redundancy is avoidable.

A back-of-the-envelope. Take a RAG call with 6,000 tokens of retrieved context and a 200-token question. The context is nearly the whole prefill. Squeeze it to 8% retention (about 480 tokens) at a near-lossless setting and you have already cut the context 12.5×. Cache that compressed, static preamble at the usual 0.1× cache-read rate, batch the non-urgent calls at 0.5×, and the discounts multiply—about 70× off the effective context cost versus resending the raw text every time (Fig. 6b). Caching and batching make tokens cheaper; compression makes them fewer. You want both.

Why the input side, and why a continuous operator. Most token-reduction work was built for vision transformers or image diffusion, where the target is FLOPs and the knob is a discrete per-layer merge count. Two things pull us elsewhere. We compress the input rather than the decode, because the prefill carries no autoregressive dependency to respect and is exactly where RAG redundancy piles up. And we reach for a continuous operator instead of a counter: treat the N context tokens as nodes of a similarity graph and let their embeddings flow under the graph heat equation $\dot{U} = -\mathcal{L}U$. Heat smooths in a predictable order, knocking out the highest graph frequencies first, so after a short time τ every token has been nudged toward the average of its neighbors. A token that moves a lot was carrying something its neighbors were not.

Contributions.

1. **Method (§4).** HKCC: a heat-residual saliency score, a diffused multiplicity-weighted merge, and a single spectral compression dial.

2. **Analysis (§4).** A convex-hull/maximum-principle safety guarantee (Prop. 2); a short-time identity equating the residual with local graph-Dirichlet energy (Prop. 1); a threshold-mode dial (Prop. 3); and a linear-time Chebyshev realization with an explicit complexity budget (Table 1).

3. **A dual allocation principle (§5).** "Budget-as-heat": a Laplacian consensus flow that reaches the equimarginal (water-filling) optimum, conservation-preserving, decentralized, and converging at rate $\lambda_2(L_G)$ (Prop. 4).

4. **Evidence (§6).** Ten experiments; notably, the finding that the diffused merge—not the saliency score—is decisive (Table 3), a broad robustness plateau, measured linear-time scaling, and a multiplicative cost model.

None of the ingredients is new on its own (§2): graph diffusion, heat kernels, and token merging all have deep literatures. What we claim is the combination, pointed at a target those literatures mostly ignore—the input-token cost of an autoregressive LLM—with the continuous dial, the free residual, and the convex-hull guarantee as the parts that distinguish it.

2 Related Work

Token merging and pruning. Token Merging (ToMe) accelerates vision transformers by merging the most similar tokens via bipartite matching with no retraining [1]; ToMeSD extends it to image diffusion with an unmerge step [2]. Spectrum-preserving variants align the merge with the token-graph spectrum [3], and attention-graph methods such as Zero-TPrune score importance from the attention graph [4]. These target model FLOPs (chiefly in vision) with a discrete per-block merge count. HKCC instead compresses the LLM *input* for API-token cost, uses a continuous diffusion time with a closed-form spectral meaning, and obtains the saliency score as a diffusion byproduct.

Prompt and context compression. Prompt compression shortens the input before the model sees it: LLMLingua drops low-information tokens guided by a small language model [5]; Gist tokens and AutoCompressors learn to summarize prompts into soft tokens [6, 7]. These are either learned or deletion-based; HKCC is training-free and information-preserving by construction, re-expressing rather than deleting redundant mass.

Decode-side / KV-cache reduction. A complementary line compresses the key–value cache during generation, e.g. heavy-hitter eviction [8]. HKCC is orthogonal and composes with such methods.

Graph diffusion and spectral filtering. Heat kernels on graphs are classical [9, 11]; diffusion maps use the same operator for manifold learning [10]; and graph signal processing formalizes spectral filters and their fast polynomial approximations [14, 12, 13]. We import the Chebyshev machinery for linear-time diffusion and the Markov-semigroup maximum principle for the safety guarantee.

3 Background

Notation. The context is $X = [x_1, \dots, x_N]^\top \in \mathbb{R}^{N\times d}$. We build a sparse symmetric affinity $W_{ij} = \exp(-\|x_i - x_j\|^2/\sigma^2)$ for $j \in \mathrm{kNN}(i)$ (symmetrized), $W_{ii} = 0$, with degree $D = \mathrm{diag}(W\mathbf{1})$. Define the random-walk operator $P = D^{-1}W$ and the random-walk Laplacian $\mathcal{L} = I - P$. The symmetric normalized Laplacian $L_{\mathrm{sym}} = I - D^{-1/2}WD^{-1/2}$ is similar to $\mathcal{L}$ (via $D^{1/2}$), sharing its spectrum $0 = \lambda_1 \le \cdots \le \lambda_N < 2$ with orthonormal eigenvectors $\{\phi_k\}$, the graph Fourier basis. The graph-Dirichlet energy of a signal f is $\frac{1}{2}\sum_{ij} W_{ij}(f_i - f_j)^2 = f^\top L_{\mathrm{sym}} f$, large when f varies sharply across edges.

Graph heat semigroup. Treating each embedding coordinate as a graph signal, $\dot{U}(\tau) = -\mathcal{L}U(\tau)$, $U(0) = X$, solves to

$$U(\tau) = \mathbf{H}_\tau X, \qquad \mathbf{H}_\tau := e^{-\tau\mathcal{L}} = e^{-\tau(I-P)} = e^{-\tau}\sum_{k=0}^{\infty}\frac{\tau^k}{k!}P^k. \tag{1}$$

Spectrally, $\mathbf{H}_\tau = \sum_k e^{-\tau\lambda_k}\phi_k\phi_k^\top$: a low-pass filter with response $e^{-\tau\lambda}$, attenuating frequency $\lambda \gtrsim 1/\tau$. *Probabilistically,* $\mathbf{H}_\tau$ is a convex combination of powers of the stochastic matrix P, hence row-stochastic with nonnegative entries: each diffused token is a weighted average of original tokens.

4 Heat-Kernel Context Compression

4.1 Problem

Given $X \in \mathbb{R}^{N\times d}$ and a budget $M \ll N$, produce a compressed context $\widetilde{X} = \{(r_a, m_a)\}_{a=1}^{M}$ of M representatives with multiplicities m_a such that a downstream attention layer reading $\widetilde{X}$ produces nearly the same output as one reading X, at a fraction of the input-token cost. Only the input is compressed; the decode path is untouched.

4.2 The heat residual

Definition 1 (Heat residual). *For* $\tau > 0$, $\rho_i(\tau) = \|x_i - [\mathbf{H}_\tau X]_i\|_2 = \|[(I - \mathbf{H}_\tau)X]_i\|_2$.

The operator $I - \mathbf{H}_\tau$ has spectral response $1 - e^{-\tau\lambda}$—a *high-pass* filter—so ρ_i is the per-token magnitude of the high-frequency content of the context at node i. A token in a locally smooth region is already near its neighborhood average and barely moves (ρ_i small, redundant); a token that is high-frequency relative to its neighbors is pulled strongly toward the local mean (ρ_i large, distinctive).

Proposition 1 (Short-time limit = local Dirichlet energy). *As* $\tau \to 0^+$, $\rho_i(\tau)/\tau \to \|(\mathcal{L}X)_i\| = \big\|\sum_j P_{ij}(x_i - x_j)\big\|$.

Proof. $\mathbf{H}_\tau X = X - \tau\mathcal{L}X + O(\tau^2)$, so $x_i - [\mathbf{H}_\tau X]_i = \tau(\mathcal{L}X)_i + O(\tau^2)$; take the norm and divide by τ. The identity $(\mathcal{L}X)_i = \sum_j P_{ij}(x_i - x_j)$ follows from $\mathcal{L} = I - P$ and $\sum_j P_{ij} = 1$. □

Thus small τ ranks tokens by local graph-gradient magnitude; larger τ judges redundancy at coarser scales. The diffusion time is a genuine *scale* parameter, not merely a threshold.

4.3 Compression operator: anchors and diffused merge

The residual tells us what to *keep distinct*; it does not license *deleting* the rest, because redundant regions can carry large aggregate attention mass. HKCC keeps distinctive tokens as anchors and re-expresses the remaining mass as diffused centroids:

1. **Anchors:** $A = \text{top-}M_i\, \rho_i(\tau)$.
2. **Assignment:** each token j routes to its nearest anchor in *diffused* coordinates, $\pi(j) = \arg\min_{a\in A} \|u_j - u_a\|$, $u = \mathbf{H}_\tau X$.
3. **Representatives:** for $C_a = \{j : \pi(j) = a\}$, $r_a = \frac{1}{|C_a|}\sum_{j\in C_a} x_j$, $m_a = |C_a|$.

At inference the model attends over $\{r_a\}$ with *proportional attention*: a merged token's logit is boosted by $\log m_a$ so it stands in for the m_a originals,

$$\mathrm{Attn}(q) = \sum_a \frac{\exp\big(q^\top r_a/\sqrt{d} + \log m_a\big)}{\sum_{a'} \exp\big(q^\top r_{a'}/\sqrt{d} + \log m_{a'}\big)}\, r_a. \tag{2}$$

4.4 Properties

Proposition 2 (Convex-hull safety / no fabrication). *For every* $\tau \geq 0$, $[\mathbf{H}_\tau X]_i \in \mathrm{conv}\{x_1, \ldots, x_N\}$ *and* $r_a \in \mathrm{conv}\{x_j : j \in C_a\}$. *Hence every compressed token lies in the convex hull of the original context.*

Proof. $\mathbf{H}_\tau$ is nonnegative and row-stochastic (§3), so $[\mathbf{H}_\tau X]_i$ is a convex combination of the x_j; each r_a is an average of a subset, hence a convex combination; the convex hull is closed under both. □

A discrete maximum principle follows coordinate-wise: $\min_j x_{j,c} \leq [\mathbf{H}_\tau X]_{i,c} \leq \max_j x_{j,c}$. In regulated settings this is an audit property: a compressed token can never take an embedding value absent from the original context.

Table 1: Per-call complexity of HKCC (κ neighbors, Chebyshev order K, budget $M \ll N$).

Stage	Cost	Notes
Graph construction	$O(N\kappa d)$	approximate NN; $O(N^2 d)$ if exact
Heat diffusion (Chebyshev)	$O(KN\kappa d)$	K sparse mat-vecs
Residual + top-M selection	$O(Nd + N \log N)$	
Assignment to anchors	$O(N\kappa d)$	nearest anchor in diffused space
Centroid merge	$O(Nd)$	
Total	$O(KN\kappa d)$	linear in N for fixed κ, K

Algorithm 1 Heat-Kernel Context Compression (HKCC)

Require: context $X \in \mathbb{R}^{N\times d}$; budget M; diffusion time τ; neighbors κ; order K

1: $W \leftarrow$ symmetric κNN cosine–Gaussian affinity of X; $\quad L_{\text{sym}} \leftarrow I - D^{-1/2} W D^{-1/2}$
2: $U \leftarrow \textsc{ChebyshevHeat}(L_{\text{sym}}, X, \tau, K)$ $\qquad \triangleright$ $U \approx e^{-\tau\mathcal{L}}X$, $O(KN\kappa d)$
3: $\rho_i \leftarrow \|x_i - u_i\|$ for all i; $\quad A \leftarrow$ indices of the M largest ρ_i
4: **for** each token j **do** $\pi(j) \leftarrow \arg\min_{a\in A} \|u_j - u_a\|$
5: **end for**
6: **for** each anchor $a \in A$ **do** $C_a \leftarrow \{j : \pi(j) = a\}$; $\; r_a \leftarrow \frac{1}{|C_a|}\sum_{j\in C_a} x_j$; $\; m_a \leftarrow |C_a|$
7: **end for**
8: **return** $\{(r_a, m_a)\}_{a\in A}$ $\qquad \triangleright$ attend via Eq. (2)

Proposition 3 (Threshold-mode dial)**.** *Fix θ and keep $\{i : \rho_i(\tau) \geq \theta\}$. As $\tau \to 0^+$ the kept set is empty, and $\rho_i(\tau) \to \|x_i - \bar{x}\|$ as $\tau \to \infty$ ($\bar{x}$ the diffusion-stationary average). Over the operative range the kept fraction varies monotonically with τ (empirically, Fig. 3a).*

We state the monotonic behavior as an empirical regularity, not a global theorem: individual residuals need not be monotone in τ, but the aggregate kept fraction is monotone up to saturation in all our runs.

4.5 Linear-time computation

Forming $\mathbf{H}_\tau$ is $O(N^3)$. We never need the operator, only its action $\mathbf{H}_\tau X$ on one signal, approximated by a Chebyshev expansion of $e^{-\tau\lambda}$ on the spectral interval. With L_{sym} rescaled to $\tilde{L} = L_{\text{sym}} - I$ (spectrum in $[-1, 1]$),

$$\mathbf{H}_\tau X \approx \sum_{k=0}^{K} c_k\, T_k(\tilde{L})X, \qquad T_0 = I,\ T_1 = \tilde{L},\ T_{k+1} = 2\tilde{L}T_k - T_{k-1}, \tag{3}$$

with c_k the Chebyshev coefficients of the rescaled exponential. Each term is one sparse product $\tilde{L}\cdot(\,\cdot\,)$ costing $O(N\kappa d)$ for a κ-NN graph; the full diffusion is $O(KN\kappa d)$, *linear in N*. The coefficients decay geometrically, so at the operating scale $K \approx 20$ reaches machine precision (Fig. 3c); larger τ needs proportionally more terms but remains a handful. For very large τ, implicit Euler $(I + \Delta\tau L)U_{n+1} = U_n$ via a few conjugate-gradient steps is preferable. Table 1 collects the budget.

5 Budget-as-Heat: Allocating Tokens Across a Pipeline

Production systems are multi-stage—a classifier, a retriever, an LLM ensemble—each consuming a slice of a fixed budget B. Let stage s receive b_s and return concave, increasing value $v_s(b_s)$. We seek $\max_{b\geq 0}\sum_s v_s(b_s)$ s.t. $\sum_s b_s = B$. KKT gives the *equimarginal* (water-filling) condition: the marginal values $m_s := v_s'(b_s)$ are equal across active stages. We realize this by a heat flow on the pipeline graph G (Laplacian L_G):

$$\dot{b} = \frac{L_G\, m}{|v_s''(b_s)|} \quad \Longrightarrow \quad \dot{m} = -L_G\, m, \tag{4}$$

the second identity following from $m = v'(b)$. Equation (4) is Laplacian consensus on the marginal-value field. Two operational features: *conservation*, since $\mathbf{1}^\top L_G = 0 \Rightarrow \frac{d}{dt}\sum_s b_s = 0$; and *locality*, since $(L_G m)_s$ depends only on neighboring marginals, giving an online, decentralized controller.

Proposition 4 (Convergence rate)**.** *Linearizing* (4) *about the equimarginal fixed point, the deviation of m from consensus obeys $\|m(t) - \bar{m}\| \leq e^{-\lambda_2(L_G)\,t}\,\|m(0) - \bar{m}\|$, where λ_2 is the algebraic connectivity of G.*

Proof. Near the fixed point $\dot{m} = -L_G m$; expand $m(0)$ in the eigenbasis of L_G. The $\lambda_1 = 0$ mode is the consensus value $\bar{m}$; every other mode k decays as $e^{-\lambda_k t}$, the slowest being λ_2. □

Figure 7b confirms the rate ordering across path, star, and complete pipelines. To be clear about what is and isn't new: the fixed point is just classical water-filling. What we add is the way you get there—a decentralized heat flow—and the reading that comes with it, where marginal value plays the role of temperature and the optimum is the point where everything is the same temperature.

6 Experiments

Setup. We isolate the mechanism on a controlled synthetic context with known redundancy structure; we do not claim real-LLM results here (see §8). We draw $N = 340$ unit-norm embeddings in $\mathbb{R}^{64}$: eight tight topic clusters (40 tokens each, redundant) plus 20 sphere-spread outliers (distinctive). The graph uses $\kappa = 10$; the working diffusion time is $\tau^\star = 1/\operatorname{median}_k \lambda_k \approx 0.96$. All figures and numbers come from a single fixed-seed script.

6.1 (C1) The residual separates redundant from distinctive tokens

Figure 1 shows the residual on the diffusion-map embedding and its distribution by token type. Outliers carry 1.42× the mean residual of cluster-core tokens. The signal is soft (distributions overlap) but systematic.

6.2 (C2) The diffused merge is the load-bearing component

We measure attention-output fidelity: for 400 random queries we compare softmax-attention output over the full context against the same over the compressed context (cosine), averaged across six contexts. The result, in Figure 2 and Table 2, is lopsided. Pruning by the residual—keep the distinctive tokens, throw away the rest—is the worst thing you can do, scraping 0.34 at 8% retention, because dropping a dense cluster discards all the attention mass it carried as a group. Re-express that mass instead, the way HKCC does, and fidelity barely flinches: 0.999 at 8%, 0.9997 at 12%, well clear of random and stride. So the residual is best read not as a list of tokens to delete but as a guide to what each surviving token should stand for; the weighted centroid keeps what deletion would have thrown out.

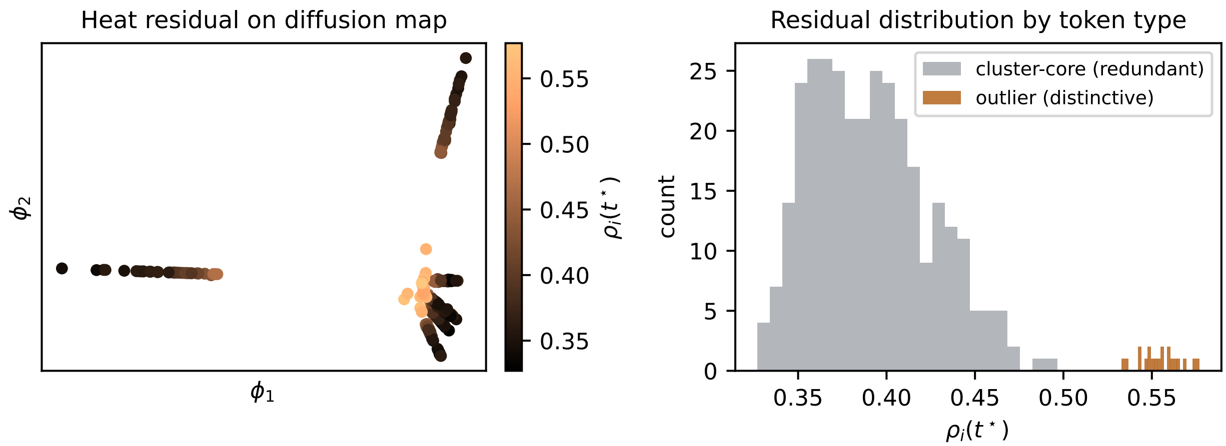


Figure 1: **Heat residual as a redundancy score.** Left: $\rho_i(\tau^\star)$ on the first two diffusion coordinates; distinctive tokens sit at the periphery with higher residual. Right: residual distributions for cluster-core (redundant) vs. outlier (distinctive) tokens (1.42× mean ratio).

Table 2: Attention-output fidelity (cosine; higher is better) at two compression budgets.

Strategy	8% kept	12% kept
Random prune	0.839	0.893
Uniform stride	0.883	0.926
Heat residual (prune only)	0.338	0.653
HKCC (residual + diffused merge)	**0.9997**	**0.9997**

6.3 What carries the method: an anchor-selection ablation

If the merge is doing the work, then which tokens we pick as anchors shouldn't matter much. It doesn't. Table 3 holds the diffused merge fixed at a 10% budget and swaps the anchor rule: random anchors, k-means centroids, and heat-residual anchors all land near 0.999, with the residual ahead only by a hair. We'd rather flag this than bury it. On clean Gaussian clusters k-means is close to optimal by construction, and once the merge re-expresses the dropped mass, the anchor rule has little left to decide. The heat formulation earns its keep elsewhere: one continuous knob τ in place of a hand-set cluster count, the residual handed to you for free, and the convex-hull guarantee. The effect that is large and robust is merge versus prune—a 0.66 fidelity gap (Table 2)—not the scorer.

6.4 (C3) The scale dial and robustness

Figure 3a shows the kept fraction varying monotonically with τ (Prop. 3); Fig. 3b shows the low-pass spectral picture; and Fig. 3c shows the Chebyshev error falling geometrically to machine precision by $K \approx 20$ at the operating scale. Figure 4a sweeps τ at fixed budget: the effect is real but modest and concentrated at aggressive budgets—at 8% retention fidelity improves from $\sim$0.9976 (very small τ) to $\sim$0.9998 for $\tau \gtrsim 1$ and stays flat over more than a decade, while at 15% retention the method is essentially τ-insensitive. Figure 4b maps fidelity over the (τ, budget) grid: HKCC occupies a broad operating plateau rather than a knife-edge optimum.

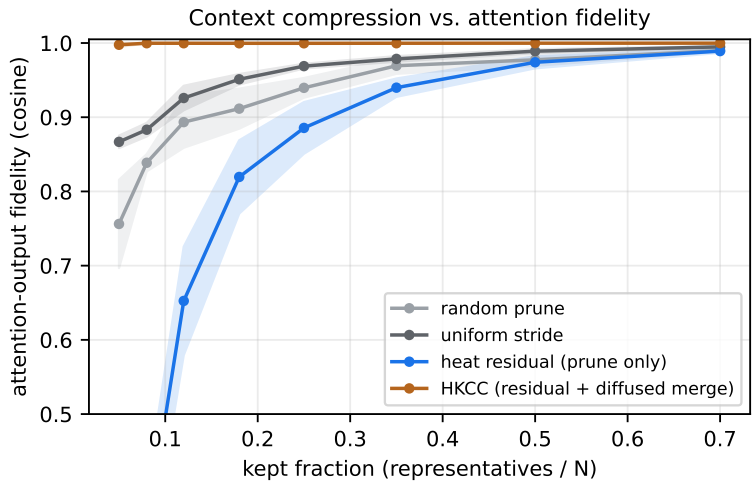


Figure 2: **Attention-output fidelity vs. kept fraction.** HKCC (orange) stays near-lossless down to very small budgets; pruning by residual alone (blue) is worst, confirming that deleting redundant mass—rather than re-expressing it—is the failure mode. Bands are ±1 s.d. over six contexts.

Table 3: Anchor-selection ablation at 10% budget, diffused merge fixed (mean ± s.d., six contexts). The merge dominates; anchor choice is a near-tie. Contrast prune-only (0.34 at 8%, Table 2).

Anchor strategy (all + diffused merge)	fidelity @ 10%
Random anchors	0.9994 ± 0.0003
k-means merge	0.9994 ± 0.0003
Heat-residual anchors (ours)	$\mathbf{0.9997} \pm 0.0001$

6.5 Why it works: the context is low-frequency

Figure 5 shows the graph-frequency spectrum of the context signal and its cumulative energy: 69.8% of the signal energy lies in the lowest 20% of graph frequencies. The context is smooth on the token graph—highly compressible—which is exactly the regime in which a low-pass diffusion operator excels.

6.6 Cost and runtime

Figure 6a times the Chebyshev diffusion against the exact matrix exponential as the context grows: the Chebyshev cost tracks the $O(N)$ reference (here 1.6 ms at N=128 to 40 ms at N=4096) while the dense exponential follows $O(N^3)$ (0.32 s already at N=1024). Figure 6b models the dollar payoff: at the 8% operating point the relative context input cost is 0.08 for compression alone, 0.028 once the static compressed preamble is cached at 0.1×, and 0.014 with batching—a $\sim$70× reduction, the levers stacking multiplicatively.

6.7 Budget-as-heat converges at rate λ_2

Figure 7a runs the flow (4) on a seven-stage path pipeline: the marginal-value spread contracts from 0.13 to 3×10^{-4}. Figure 7b varies the pipeline topology: convergence is exponential at a rate ordered exactly by algebraic connectivity (λ_2: path 0.07, star 1.00, complete 12.0), confirming Prop. 4.

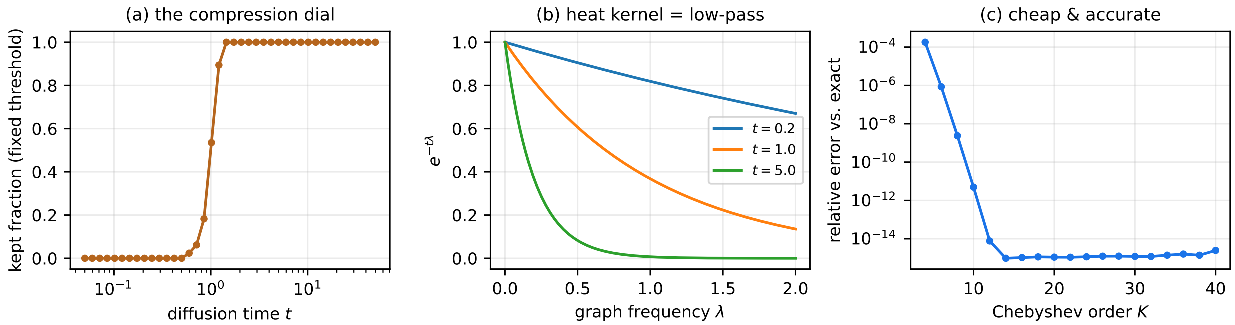
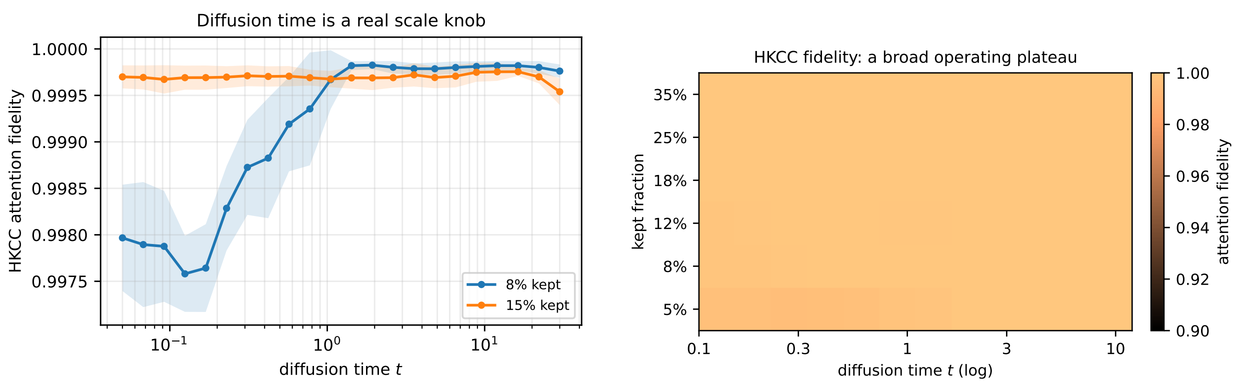


Figure 3: **The dial and its cost.** (a) at a fixed threshold the kept fraction is monotone in τ; (b) the heat kernel is a low-pass filter $e^{-\tau\lambda}$; (c) the Chebyshev approximation reaches machine precision by $K \approx 20$ at the operating scale.



(a) Fidelity vs. τ at fixed budget.

(b) Fidelity over (τ, budget).

Figure 4: **The operating region is broad.** Diffusion time has a modest, monotone effect concentrated at aggressive budgets (left), and HKCC maintains high fidelity across a wide (τ, budget) plateau (right).

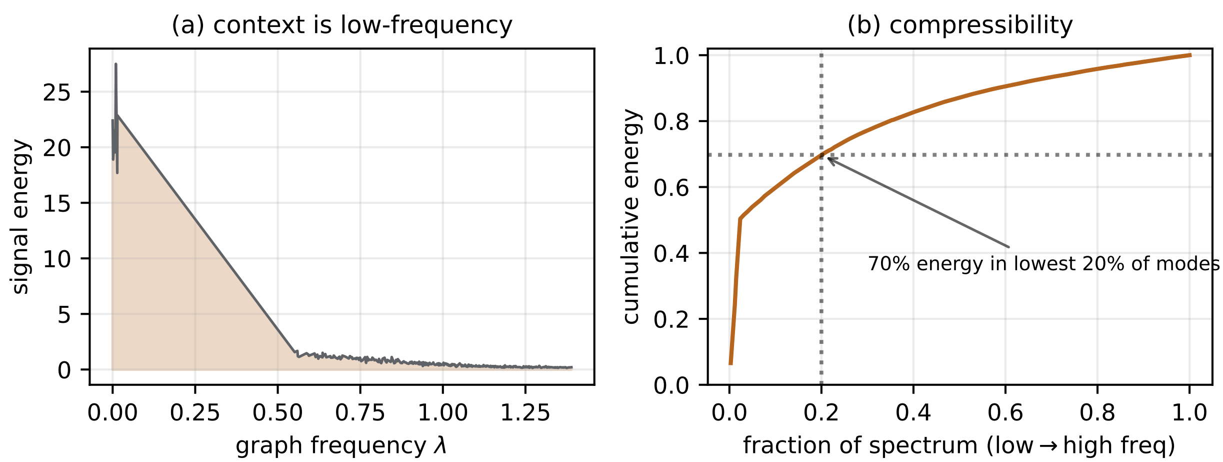
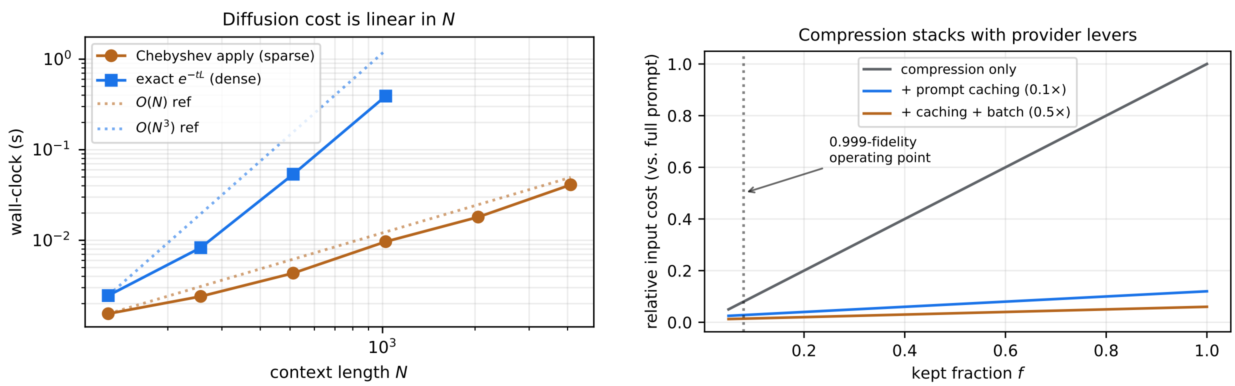


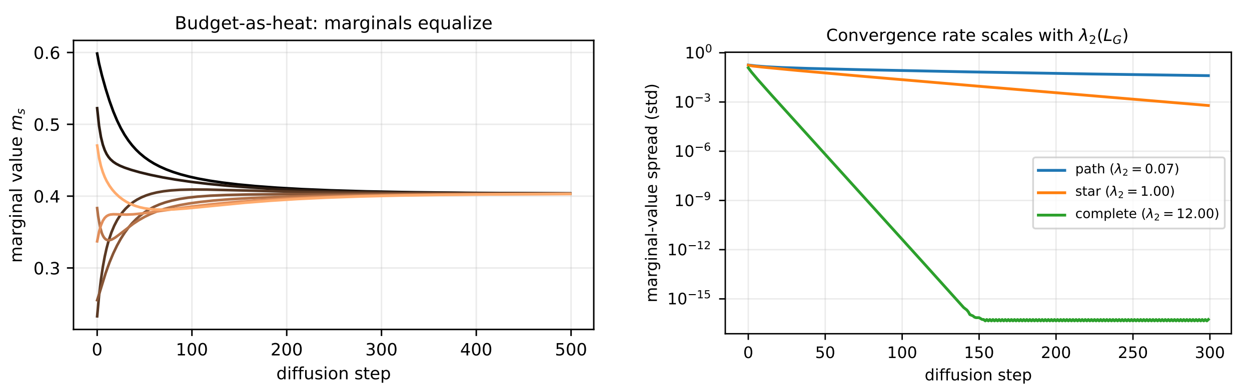
Figure 5: **Compressibility diagnostic.** (a) signal energy concentrates at low graph frequencies; (b) the lowest 20% of modes already hold $\sim$70% of the energy, so the context is intrinsically low-rank on the token graph.



(a) Measured wall-clock vs. N.

(b) Relative input cost vs. retention.

Figure 6: **Cheap to run, cheap to serve.** Left: diffusion cost is empirically linear in context length where the exact operator is cubic. Right: compression composes multiplicatively with prompt caching and batching.



(a) Marginals equalize (path pipeline).

(b) Rate scales with $\lambda_2(L_G)$.

Figure 7: **Budget-as-heat allocation.** The Laplacian consensus flow reaches the equimarginal optimum using only neighbor information (left), at a rate set by the pipeline's algebraic connectivity (right).

7 Discussion

What's actually new. Now that the ablation has spoken, the claim is easy to state. The big, repeatable effect is merge-not-prune: re-expressing redundant mass as multiplicity-weighted centroids is near-lossless exactly where deleting it is a disaster. The heat machinery is what turns that observation into a single-parameter, training-free method with some structure behind it—a continuous spectral dial (Prop. 3), a saliency score that falls out of the diffusion for free (Prop. 1), the convex-hull guarantee (Prop. 2), and a linear-time implementation (Table 1)—all of it aimed at input-token cost rather than vision FLOPs. The pieces are old; the assembly, and the target, are the point.

Deployment and caching. HKCC targets the prefill, the safe place to compress, and is a natural RAG front-end: compress retrieved chunks before they reach the model, where cross-chunk redundancy is highest. Compression changes the prefix, which interacts with prefix-based prompt caching: a *static* compressed preamble caches normally and compounds the savings (Fig. 6b), but *per-query* compression yields a fresh prefix each call and forgoes cache hits. The right pattern is to compress the stable context once, cache it, and leave only the per-query tail uncompressed.

8 Limitations and Threats to Validity

Our evidence is synthetic. The attention-output cosine is a proxy for—not a measurement of—downstream task quality; the cluster-plus-outlier generator is an idealization of real redundancy, which is messier and partly positional, and on which k-means is unusually strong; and we evaluate a single attention layer rather than a full network. We do not yet report real-LLM task metrics, end-to-end token-cost reductions, or latency. The diffusion time τ is set by a spectral heuristic rather than learned. Finally, building even a sparse κNN graph adds pre-prefill overhead that must be amortized against the tokens saved; for short contexts the break-even may not be reached.

9 Conclusion and Future Work

Heat-Kernel Context Compression scores token redundancy with the graph heat equation and folds the redundant mass into diffused, multiplicity-weighted centroids. It is training-free, has one knob, comes with a convex-hull guarantee, and runs in linear time. In ten synthetic experiments it stays near-lossless under attention down to 8–12% retention. The merge is what makes that work; the diffusion-time knob is forgiving; the cost is linear where the exact operator is cubic; and the savings ride on top of caching and batching.

The honest next step is to stop testing on synthetic data. Wire HKCC into a real RAG-QA pipeline, report exact-match and F1 against the input-token reduction it actually buys, and put it head to head with LLMLingua-style prompt compression and ToMe-style merging. After that, the open questions that interest us most are learning τ per context instead of fixing it, an anisotropic diffusion that respects word order and causality, and a single heat-flow controller that tunes per-stage compression and the budget-as-heat allocation of §5 together.

Reproducibility. Every figure and number in this paper comes from a single self-contained simulation script (synthetic data, fixed seed) accompanying it.

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